



GE Healthcare

Technical Publications

Direction 5161287-100TPH

Revision 1

Signa® DISCOVERY MR750™

(Main Disconnect Panel)

Service Manual

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Operating Documentation

GE Industrial Systems

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GE (USA) CONFIDENTIAL AND PROPRIETARY

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Troubleshooting and servicing should be performed only by a qualified electrician.



AC and DC control power is present when the breakers are in the open position. De-energize incoming power and lockout before servicing this panel.

This panel has been wired for use at 480V/60HZ, the panel can be converted for use at 380-400-415V/50-60HZ by changing the two transformer primary wiring connections from 480V to 400V. Refer to Page 10.

1- Theory of operation

This panel incorporates a number of features desirable by MR installations to minimize down time, protect PDU (Power Distribution Unit) electronics, reduce operating costs, and reduce operational delays after a power outage. The panel comes wired with a single main circuit breaker and individual branch breakers for the PDU and Heat Exchange Cabinet (HEC).

The PDU branch circuit is controlled by a solid state circuit breaker with undervoltage trip, which opens the PDU circuit breaker on any loss of power. The PDU under-voltage trip circuit breaker CB2, protects the PDU's sensitive electronic equipment from sags and surges which immediately follow power loss from blackouts, storms, utility reclosure operations and out of phase automatic transfer switch operations by tripping the PDU circuit breaker on any loss of power. The Heat Exchange Cabinet (HEC) branch circuit utilizes an automatic restarting control circuit, which restores power after a power outage. A 12V DC relay CR1 controls the operation of the HEC branch circuit contactor. Operation of the 12V DC battery charger and battery provide the control power for relay CR1. Relay CR1 must be energized in order for the PDU circuit breaker under-voltage trip or the HEC branch circuit contactor, C1 to close. The panel has a HEC contactor, which is controlled by the DC autostart control circuit. DC battery control circuit protection time is based upon the condition of the battery but is expected to hold for at least 48 hours. The DC Battery has a life of 5 years at which time the battery and charger must be replaced to maintain auto-restart capability.

PDU and HEC power Emergency Off disconnection is also provided by this panel. Two normally closed remote emergency off pushbuttons are included with this panel and the cover-mounted power off pushbutton provide immediate shutdown for the entire system. If required for a particular installation, additional remote emergency off pushbuttons may be ordered from GE Supply 1-800-200-9760. Remote emergency off pushbuttons are listed at the end of this manual. Restoration of power after an emergency off operation is accomplished by first pressing the System ON pushbutton on the cover of the panel and subsequently resetting the PDU circuit breaker by moving the PDU circuit breaker handle to OFF then to ON. Circuit breaker CB3 provides power for the HEC as indicated on the attached wiring diagram.

The battery charger red indicating light is constantly illuminated while the charging operation is in progress and begins to dim and flicker when charging operation is complete. However, NO red indicating light indicates either a loss of 120V or charger failure.

SPECIFICATIONS

ELECTRICAL SPECIFICATIONS

UL design and ratings require the application of wire at the 75-degree ampacity ratings of NEC table 310.16. Wire rated at higher temperature ratings such as 90 degree may be used but ONLY at the 75-degree ampacity of NEC Table 310.16.

UL, cUL labeled and CE labeled.

3 Phase, 4 Wire + Ground

Main Feed wire sizes;

 Main circuit breaker #8 - 350KCMIL maximum.

 Main Ground Bus #6-250KCMIL maximum.

 Main Neutral Bus #4 – 350KCMIL maximum.

480V/60HZ, 380-400V/50HZ

Panel Short Circuit Current Rating is 25,000A @ 480V

Main fuses are provided for Short Circuit Current protection. These fuses are oversized at 200 amperes for the PDU circuit and 60 ampere for the HEC cabinet to provide back-up protection.

Circuit Breaker CB2 utilizes 200A rated fuses for the PDU

Circuit Breaker CB3 utilizes 60A rated fuses for the Heat Exchanger cabinet

380, 400, & 415Volt, 50/60HZ

Circuit Breakers CB1, CB2 and CB3, UL 489 & CSA C 22.2

Circuit Breaker Interrupting Rating 25,000 AIC Symmetrical 50/60 Hertz

Panel Short Circuit Current Rating is 25,000 Amperes RMS symmetrical at 480 volt AC.

IEC 947-2 Interruption Capacity

SFHA Main Circuit Breaker 380-415V, Ue, Icu 35,000 RMS Symmetrical AMPS 50/60 Hertz.

SFHA Main Circuit Breaker 380-415V, Ue, Ics 17,000 RMS Symmetrical AMPS 50/60 Hertz.

SEHA Branch Circuit breakers 380-415V, Ue, Icu 15,000 RMS Symmetrical AMPS, 50/60 Hertz

SEHA Branch Circuit breakers 380-415V, Ue, Ics 10,000 RMS Symmetrical AMPS, 50/60 Hertz

ENVIRONMENTAL SPECIFICATIONS

FOR INDOOR USE ONLY

 Temperature 59-95F (15-35C)

 Humidity 30-75%, NON-CONDENSING

COVER WARNING LABELS

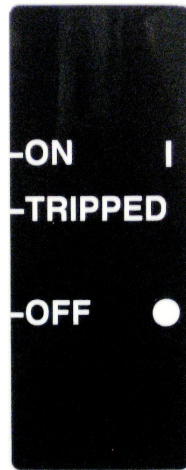


Figure 1-1-3-1
Circuit Breaker plate



Figure 1-1-3-2
Pushbutton plates

WARNING: THIS HEAT EXCHANGE CABINET IS FED BY AN AUTOMATIC RESTART CIRCUIT WHICH ENERGIZES THE SYSTEM HEAT EXCHANGE CABINET AUTOMATICALLY AFTER ANY POWER INTERRUPTION.

Figure 1-1-3-3
Warning label

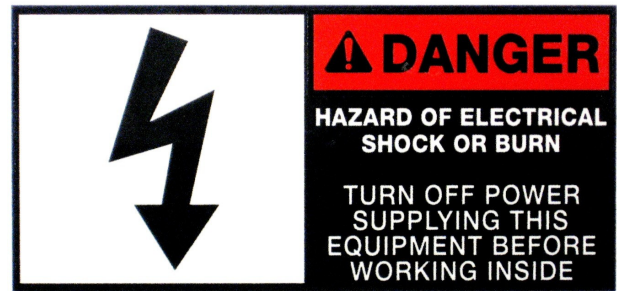


Figure 1-1-3-4
Danger shock label

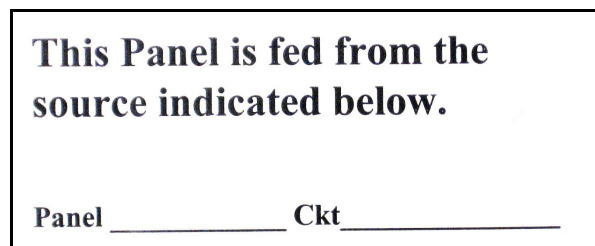


Figure 1-1-3-5
Source label

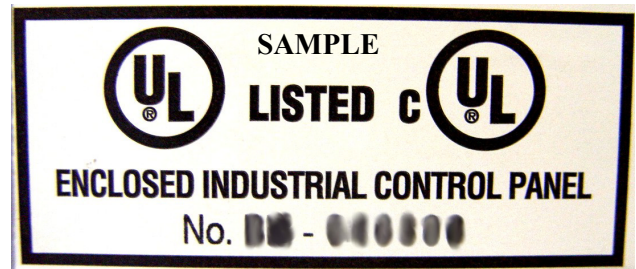


Figure 1-1-3-6
Nationally recognized test Laboratory approval label

1-2 Panel Dimensions

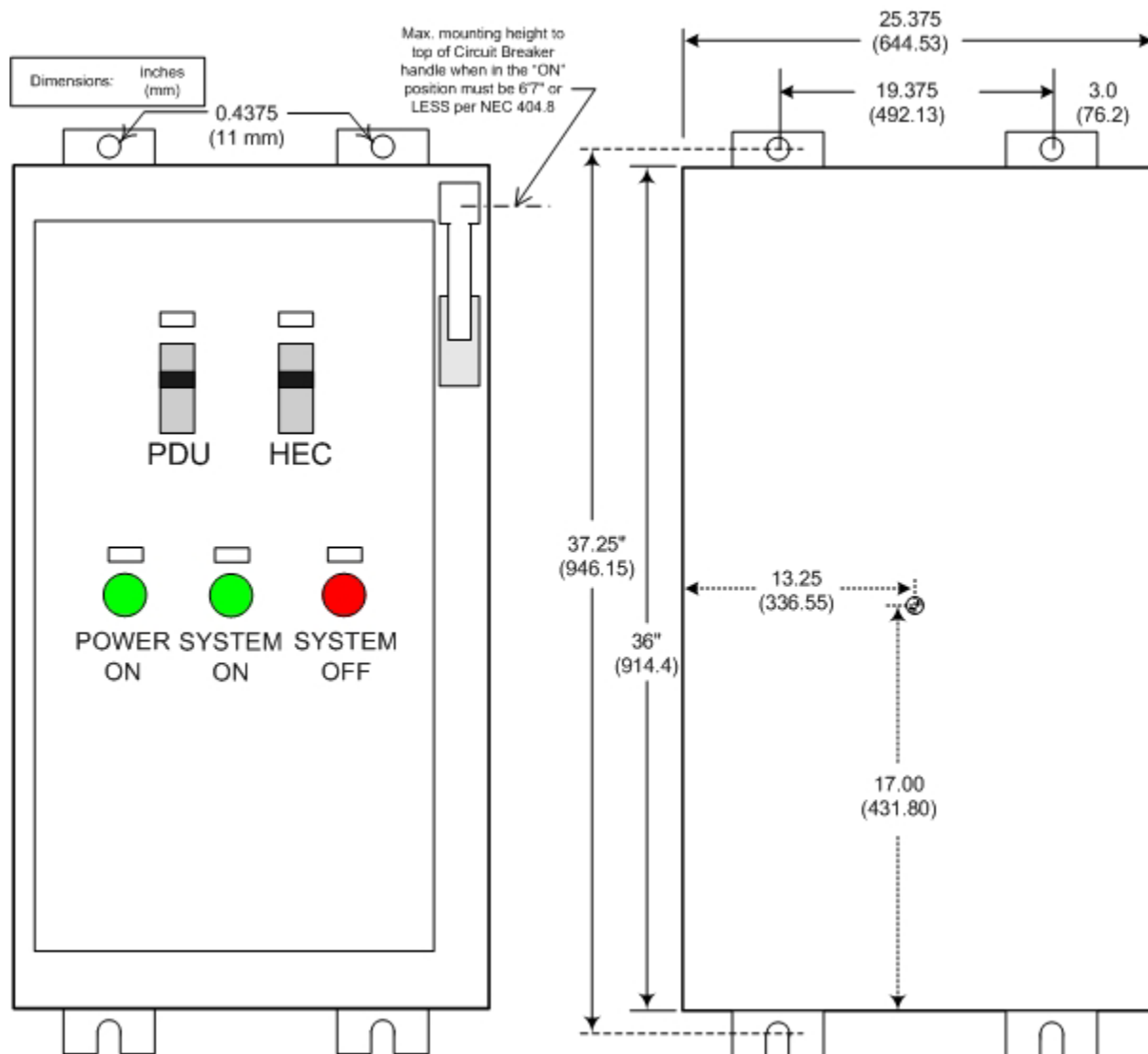
Refer to Table 1-1 and Illustration 1-1 for the physical dimensions of the panel. The panel may be recessed approximately 6.95 inches (176.5 mm) into the wall for semi-flush installations. See Illustration 1-1 for the front view and mounting pattern of the Main Disconnect Panel.

TABLE 1-1
 PANEL EXTERNAL PHYSICAL DIMENSIONS

PARAMETER	DIMENSIONS	
Height-Enclosure	36.0 inches	914.4 mm
To edge of mounting flanges	38.5 inches	977.9 mm
Width	25.375 inches	644.525 mm
Depth	9.25 inches	234.54 mm
Weight	163 lbs.	73.9 kg
Door Swing Radius	19.75 inches	501.65 mm
Center of Gravity as indicated below and next page.		

FRONT VIEW

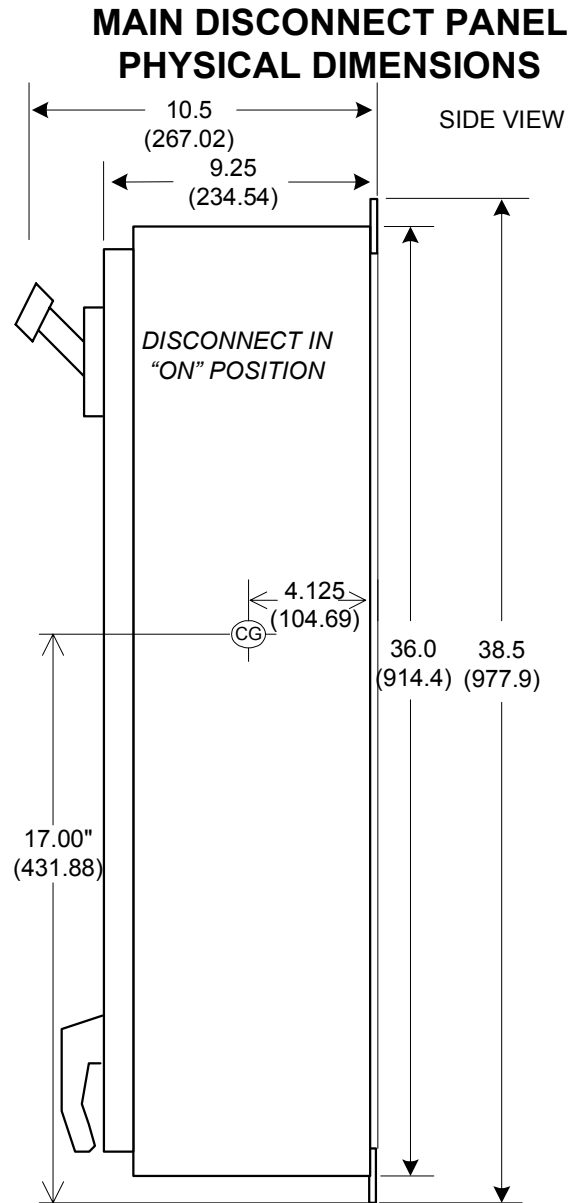
MOUNTING PATTERN



FRONT AND BACK VIEW OF MAIN DISCONNECT PANEL
 ILLUSTRATION 1-1

1-3 PANEL SIDE VIEW PHYSICAL DRAWING

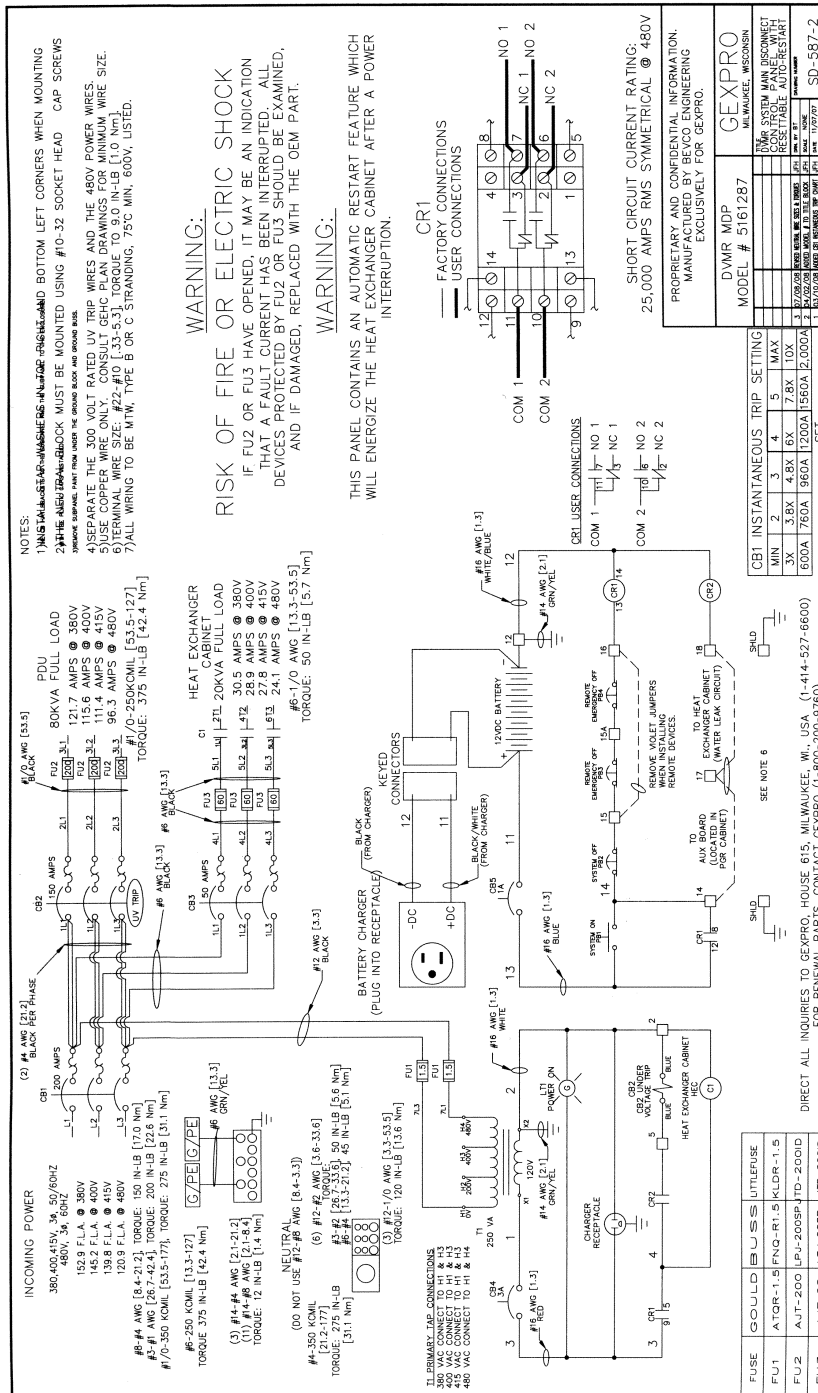
See Illustration 1-2 for panel side view physical drawing.



ALL DIMENSIONS INCHES (MILLIMETERS)
PRINT IS NOT TO SCALE
CONDUIT AREA MUST BE FIELD VERIFIED

SIDE VIEW PHYSICAL DRAWING
ILLUSTRATION 1-2

1-4 Electrical Wiring Diagram



1-5 Warning Labels

Warning labels shown in this section are provided with the MDP and are to be affixed to the Heat Exchanger cabinet as indicated.

1-5-1 Heat Exchange Cabinet label

Two HEC labels are provided with the MDP. Illustration 1-6 shows the HEC warning label, which is to be affixed to the front panels of the HEC cabinet.

**WARNING: THIS HEAT
EXCHANGE CABINET IS FED BY AN
AUTOMATIC RESTART CIRCUIT,
WHICH ENERGIZES THE SYSTEM
HEAT EXCHANGE CABINET
AUTOMATICALLY AFTER ANY
POWER INTERRUPTION.**

HEC WARNING LABEL
ILLUSTRATION 1-6

2 INSTALLATION

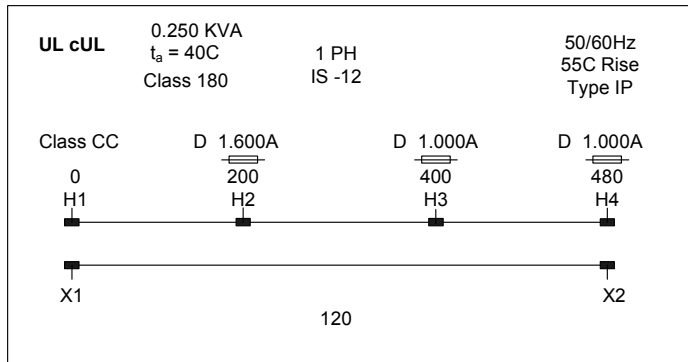
2-1 Installation

- Incoming power is connected to the main breaker located at the top of the Main Disconnect Panel. Maximum conductor size is #8-350KCMIL. If larger conductors are utilized then the larger conductors are terminated in a junction box above the MDP and a 4/0 - 350KCMIL conductors are spliced in the junction box and terminated to the Main CB, CB1. The 200A rating of the upstream overcurrent device will protect the 4/0- 350KCMIL conductors, which are rated 310 amperes as well as the larger conductors.
- There are no knockouts in the panel for conduits so the installer can punch the knockouts where they prefer to have them.
- All conductors must be sized per the GEHC wiring tables on the installation drawings or the GEHC system installation manual. The PDU is connected to the load side of fuse block FU2, terminals 3L1, 3L2, and 3L3. Conductor range is #6 AWG to 250KCMIL conductors.
- Heat Exchange Cabinet (HEC) connections are made at contactor C1 terminals 2T1, 4T2, and 6T3. Conductor size is #6AWG- 1/0 AWG.
- Verify that the DC battery charger is plugged into the proper receptacle and that the DC output voltage is greater than 12V (12 to 13.65V).
- Mounting height of the centerline of the main breaker or the handle of the operating mechanism when in the ON position must not exceed 6ft-7in. (2m) above the floor per National Electric Code (NEC # 404.8).
- Complete the label on the cover of the panel indicating the location and circuit of the power source providing power to this panel.
- Two labels are supplied indicating "Warning of automatic restart OF THE HEC" must be installed on the HEAT EXCHANGE CABINET. See section 1-5 "Warning Labels".

The control circuit wiring must be completed by installing the two remote emergency off pushbuttons as shown on the wiring diagrams on the inside of the door. The PDU circuit breaker and HEC contactor will not operate properly unless the remote emergency off pushbuttons are installed and the factory installed temporary jumper removed.

♦ **For warranty parts or assistance contact GEXPRO (Formerly GE SUPPLY) – Milwaukee, WI (414) 527-6600, CST.**

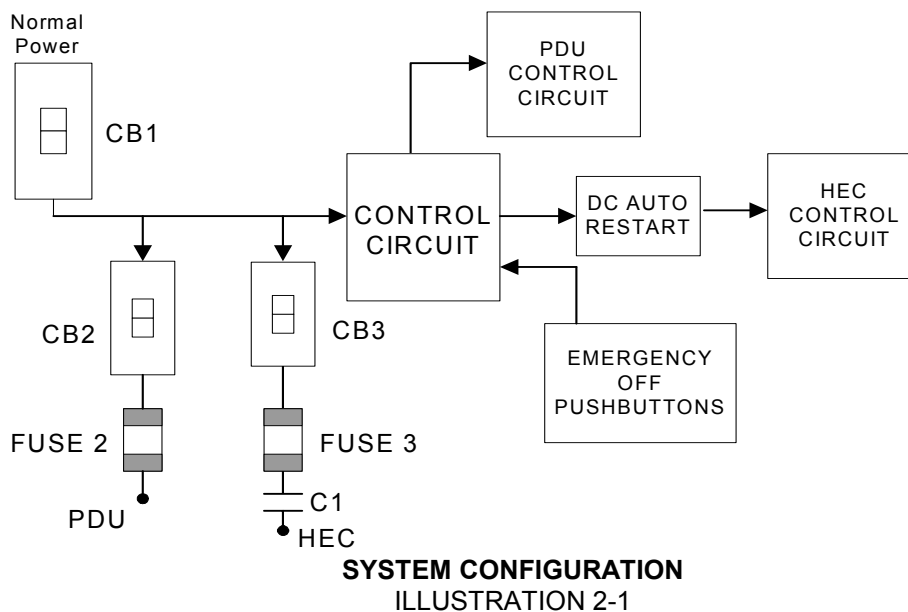
2- 2 250VA Control Transformer CPT1 480V to 400V Conversion



1. Open two-pole fuse holder FU1.
 2. Remove wire originally connected to transformer terminal H4.
 3. Connect wire from Step 2 to transformer terminal H3.
 4. Close fuse holder FU1.
- NOTE: Transformer 200V tap, H2, is not used in this application.

2-3 SYSTEM CONFIGURATION

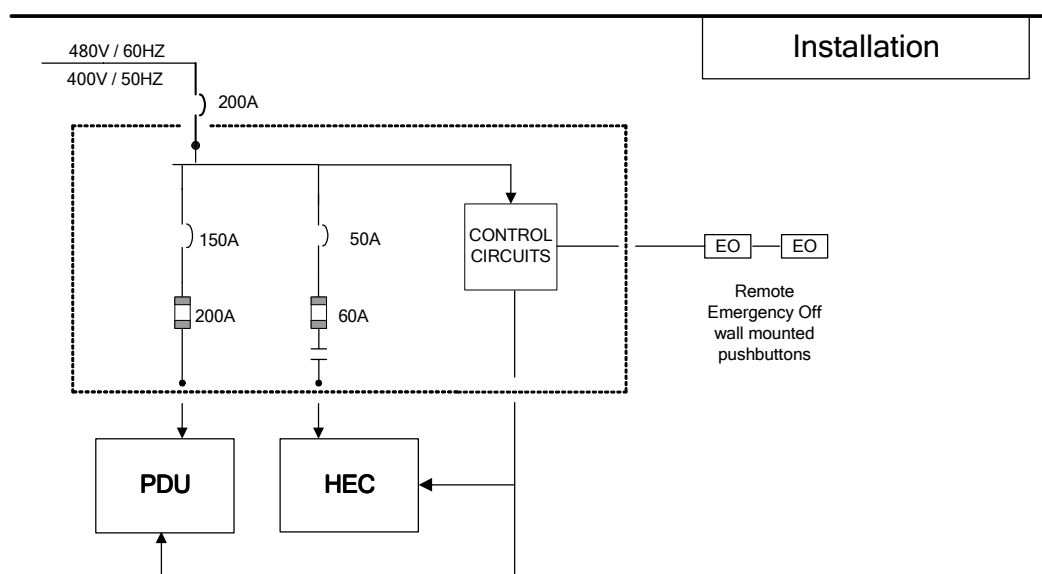
See illustration 2-1 for the System Configuration.



2-4 INCOMING POWER CONFIGURATION

See illustration 2-2 for the incoming power configuration.

INCOMING POWER CONFIGURATION
ILLUSTRATION 2-2



- Fuses are oversized for back-up short circuit protection.

2-5 TESTING PANELS

TESTING ACTION

1. Close Main Circuit Breaker, CB1. Close Heat Exchanger circuit breaker CB3.
2. Press the System ON pushbutton and then reset PDU circuit breaker CB2. Verify the (HEC) Heat Exchanger Cabinet contactor C1 is ON.
3. Press any emergency off pushbutton.
4. Press the System ON pushbutton and then reset PDU circuit breaker CB2. Verify the (HEC) Heat Exchanger Cabinet contactor C1 is ON.
5. Press any emergency off pushbutton.

VERIFY

1. Green Power On light should light indicate system control power is available.
2. Verify with voltmeter that three-phase power is available for the PDU at Fuse FU2 terminals 3L1, 3L2, 3L3 and that Heat Exchanger Cabinet contactor C1 is closed. Verify three-phase power at contactor C1 terminals 2T1, 4T2, and 6T3. Green pilot light should be on indicating control power is available.
3. PDU circuit breaker CB2 must trip (open) and HEC contactor C1 must de-energize and stay de-energized. Verify with a voltmeter that the HEC power at the contactor C1 terminals 2T1, 4T2, and 6T3 is disconnected and the PDU power at Fuse FU2 lugs 3L1, 3L2, and 3L3 is disconnected at the fuse load terminals.
4. Verify with voltmeter that three-phase power is available for the PDU at Fuse FU2 terminals 3L1, 3L2, 3L3 and that Heat Exchanger Cabinet contactor C1 is closed. Verify three-phase power at contactor C1 terminals 2T1, 4T2, and 6T3. Green pilot light should be on indicating control power is available.
5. PDU circuit breaker CB2 must trip open and HEC cabinet contactor C1 opens.

TESTING ACTION

VERIFY

- | | |
|---|---|
| <p>6. Starting with the de-energized state of #5, turn OFF incoming power at the panel main breaker CB1, feeding power to this panel.</p> <p>7. With incoming power OFF from step 6, now restore main power to panel by turning panel main circuit breaker CB1 ON (switch to ON).</p> <p>8. Press the System ON pushbutton and then reset PDU circuit breaker CB2. Verify the (HEC) Heat Exchanger Cabinet contactor C1 is ON and the green power on light is ON.</p> <p>9. Press the System ON pushbutton. Verify the PDU circuit breaker is closed. Turn OFF main circuit breaker CB1 power <u>and then turn on the main circuit breaker CB1 to restore power to the panel.</u></p> | <p>6. The green light must be OFF indicating loss of control power.</p> <p>7. Upon restoration of power the HEC contactor C1 must remain OFF. This demonstrates that the emergency off function remains disabled during a loss of power. There should be no auto restart after any emergency OFF operation. The PDU circuit breaker was tripped in Step 5 and remains tripped.</p> <p>8. The HEC contactor can be heard closing when the system ON push-button is pushed.</p> <p>9. The green pilot light must be ON upon restoration of power. The PDU circuit breaker and HEC contactor should turn OFF with loss of power caused by opening main circuit breaker CB1. Only the HEC contactor C1 must turn ON with the restoration of power indicating the proper operation of the automatic restart feature. The PDU circuit breaker CB2 turns off on any loss of power and remains off with restoration of power. PDU CB2 opens and does not close upon power restoration. The PDU does not automatically restart after a power loss. This demonstrates the DC supervisory circuit is operating properly and the HEC auto restart feature is operational.</p> |
|---|---|

WARNING!

AC AND DC CONTROL POWER IS PRESENT WHEN THE BREAKERS ARE IN THE OPEN POSITION. MAKE SURE NO ONE IS WORKING ON THE EQUIPMENT, WHICH THIS PANEL FEEDS, PRIOR TO TESTING THE PANEL.

3-TROUBLESHOOTING

Auto Restart function, PDU, and the HEC contactor will not operate properly unless the remote emergency off pushbuttons are wired as indicated on the wiring diagram and the factory installed temporary jumpers are removed.

Make sure the remote emergency off pushbuttons are wired through normally closed contacts.

Operation of any emergency off pushbutton should de-energize relay CR1, shutting the entire system down.

Check for broken or burnt out LED pilot lights.

Check to see that battery charger is firmly installed in its receptacle and 12VDC wired to the battery. 12V DC output voltage must be 12 to 13.65V DC under normal operating conditions. If DC voltage is less than 11V, replace the charger and battery. 12V DC relay CR1 controls the operation of all branch circuit contactors. Operation of the 12V DC battery charger and battery provide the control power for relay CR1. Relay CR1 must be energized in order for any branch circuit contactor to close.

The battery charger red indicating light is constantly illuminated while the charging operation is in progress and begins to dim and flicker when charging operation is complete. However, NO red indicating light indicates either a loss of 120V or charger failure

NOTES:

***FOR WARRANTY PARTS ASSISTANCE CONTACT:
GEXPRO – MILWAUKEE, WI. (414)-527-6600 CST***

4- MAINTENANCE

4-1 General Maintenance

Note

The auto restart circuit requires the 7 Ah DC battery to be functional. Semi-annually verify that the 12V DC charging circuit is operational.

Battery and Charger Test

1. Open main circuit breaker and press the system on pushbutton to energize CR1 and CR2.
2. Measure the 12-13.8VDC voltage at the battery quick connect terminals with the battery loaded. IF voltage is less than 11 volts or rapidly decreases battery replacement and charger replacement is necessary.
3. Turn ON the main circuit breaker. The charger should indicate it is on. Measure DC voltage of charger at the battery quick connect terminals.
DC voltage must be between 12 and 13.65 volts.

Normal operation of DC circuit is from the DC charger with battery fully charged in float stage. If battery is suspect or questionable, replace immediately as this is an insignificant maintenance cost. The manufacturer of the battery is Powersonic Battery and the manufacturer for the charger is EDS. Contact GEXPRO (Formally GE Supply) for battery or charger replacement.

Battery# PS1270
Charger# EDS 11027

Specific battery maintenance and recharge/discharge information can be found in attached Powersonic documents and on the Powersonic web site (below).

Battery life is 5 years at which time it must be replaced. Initial date code on top of battery is read as (mm/dd/y/c). Where c = Q.C. inspector code.

Powersonic Battery can be located at: www.power-sonic.com.

GEXPRO (Formally GE Supply)
12221 W. Feerick St.
Milwaukee, WI 53222
Phone 414-527-6600
Fax 414-527-6652

GEXPRO (Formally GE Supply)
National Service Center
7602 Woodland Drive
Indianapolis, IN 46268
Phone 1-800-243-7313

See illustration 4-1 and 4-2 for battery manufacturer's data sheet.

Battery Data

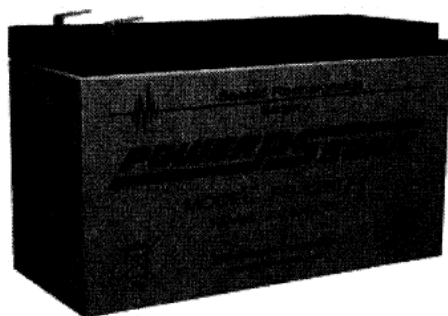


POWER SONIC

Rechargeable Sealed Lead-Acid Battery

PS-1270

12 Volt 7.0 Amp. Hrs.



Features:

- Absorbent Glass Mat (AGM) technology for superior performance.
- Valve regulated, spill proof construction allows safe operation in any position.
- Power/volume ratio yielding unrivaled energy density.
- Rugged plastic case and cover
- Approved for transport by air. D.O.T., I.A.T.A., F.A.A. and C.A.B. certified.
- U.L. recognized under file number MH 20845.



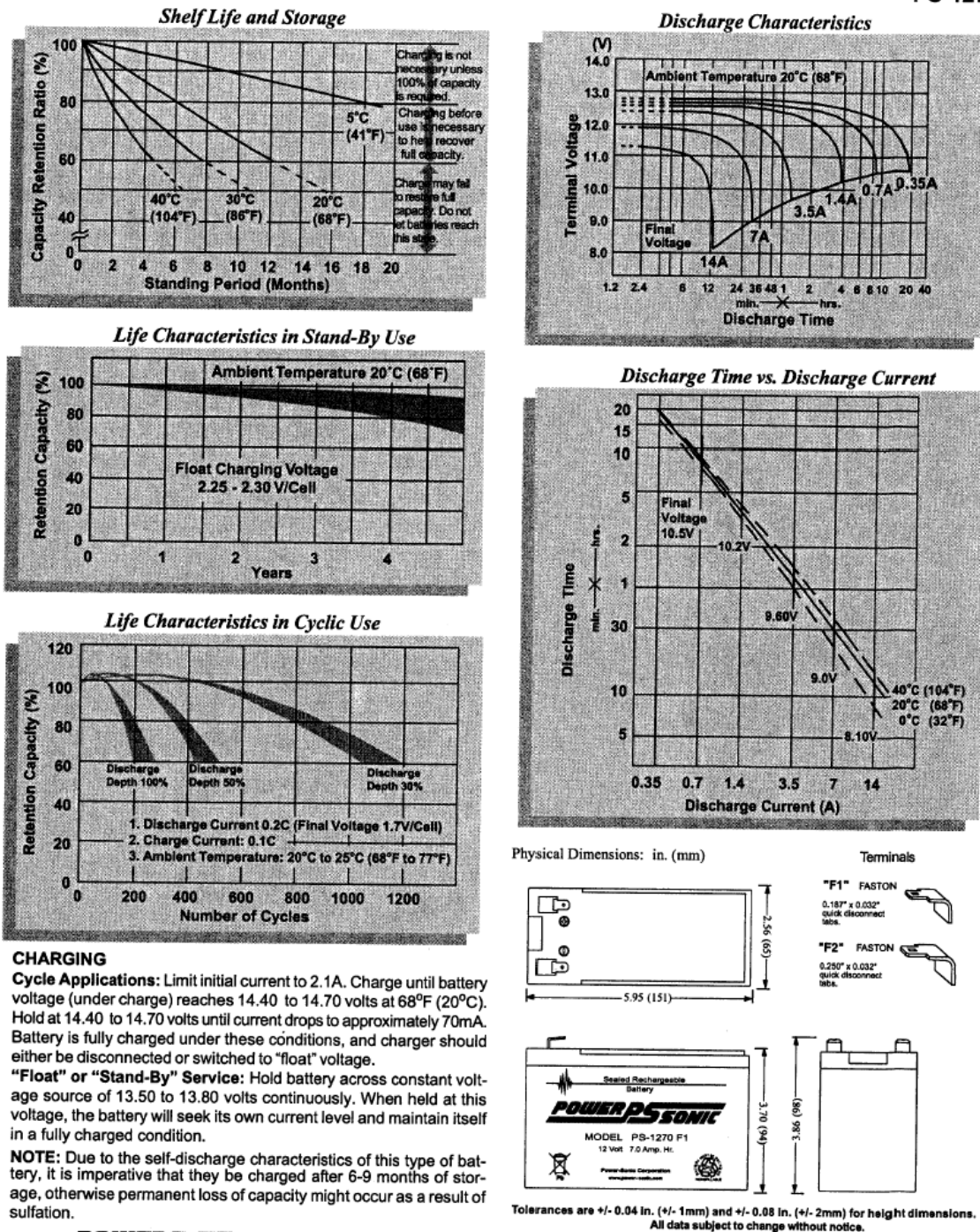
PERFORMANCE SPECIFICATIONS

Nominal Voltage.....	12 volts (6 cells in series)
Nominal Capacity	
20 hour rate (350mA to 10.50 volts)	7.0 A.H.
10 hour rate (850mA to 10.50 volts)	6.5 A.H.
5 hour rate (1.2A to 10.20 volts)	6.0 A.H.
1 hour rate (4.5A to 9.00 volts)	4.5 A.H.
15 min rate (14A to 9.00 volts)	3.5 A.H.
Approximate Weight.....	5.70 pounds (2.59 kg)
Energy Density (20 hour rate).....	1.43 Watt-hours/cubic inch (87.3 Watt-hours/l)
Specific Energy (20 hour rate).....	14.7 Watt-hours/pound (32.4 Watt-hours/kg)
Internal Resistance (Fully Charged Battery).....	22 milliohms (approximately)
Maximum Discharge Current (≤ 7 Min.).....	21 amperes
Maximum Short-Duration Discharge Current (≤ 10 Sec.).....	70 amperes
Terminal configurations	Quick disconnect tabs, "F1": 0.187" x 0.032", mate with AMP. INC. FASTON "187"
	Quick disconnect tabs, "F2": 0.250" x 0.032", mate with AMP. INC. FASTON "250"
Shelf Life — % of nominal capacity at 68° F (20° C)	
1 Month.....	97%
3 Months.....	91%
6 Months.....	83%
Operating Temperature Range	
Charge.....	-4°F (-20°C) to 122°F (50°C)
Discharge.....	-4°F (-20°C) to 140°F (60°C)
Case	ABS Plastic

BATTERY MANUFACTURER DATA SHEET

ILLUSTRATION 4-1

TECHNICAL DATA: _____ **PS-1270**



POWERPS-SONIC SALES & MARKETING 3106 Spring Street Redwood City, CA 94063 USA Tel: 650-364-5001 Fax: 650-366-3662 national-sales@power-sonic.com		ISO9001-2000 FM39170	CUSTOMER SERVICE 7550 Panasonic Way San Diego, CA 92154 USA Tel: 619-661-2030 Fax: 619-661-3648 customer-service@power-sonic.com
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BATTERY MANUFACTURER DATA SHEET
 ILLUSTRATION 4-2

4-2 Battery Manufacturer's Material Safety Data Sheet



MATERIAL SAFETY DATA SHEET

PS, PSH, PSG, PG and Power Sport Series
Valve Regulated (VRLA) Batteries
Absorbed Electrolyte (AGM)

Section 1 - Product Identification

Manufacturers Name Power-Sonic Corporation, 7550 Panasonic Way San Diego, CA 92154	Emergency Telephone Numbers: CHEMTREC (Domestic): (800) 424-9300 CHEMTREC (International): (703) 527-3887
	Telephone Number for Information Power-Sonic Corporation: (619) 661-2020
	Date Issued: January 2, 2007
<i>The information contained within is provided as a service to our customers and is for their information only. The information and recommendations set forth herein are made in good faith and are believed to be accurate at the date compiled. Power-Sonic Corporation makes no warranty expressed or implied.</i>	

Section 2 - Hazardous Ingredients/Identity Information

Components	CAS Number	Approx Wt. %	OSHA PEL (µg/m³)	ACGIH TLV (µ/m³)	NIOSH (µ/m³)
Inorganic Lead/Lead Compounds	7439-92-1	65%-75%	50	150	10
Tin	7440-31-5	<0.5%	2000	2000	N/A
Calcium	7440-70-2	<0.1%	N/A	N/A	N/A
Electrolyte: Dilute sulfuric Acid	7664-93-9	14-20%	1000	1000	1000
Fiberglass Separator	-	5%	N/A	N/A	N/A
Case Material: Acrylonitrile Butadiene Styrene (ABS)	9003-56-9	5-10%	N/A	N/A	N/A

Inorganic lead and electrolyte (sulfuric acid) are the main components of every Valve Regulated Lead Acid battery supplied by Power-Sonic Corporation. Other ingredients may be present dependent upon the specific battery type. For additional information contact Power-Sonic Corporation Technical Department.

Section 3 - Physical/Chemical Characteristics

Components	Density	Melting Points	Solubility (H2O)	Odor	Appearance
Lead	11.34	621 F°	None	None	Silver-Gray
Lead Sulfate	6.20	1950 F°	40mg/l(60 F°)	None	White Powder
Lead Dioxide	9.40	554 F°	None	None	Brown Powder
Sulfuric Acid	About 1.30	203-240 F°	100%	Sharp penetrating pungent	Clear Colorless Liquid
Fiberglass Separator	N/A	N/A	Slight	None	White Fibrous
Case Material: Acrylonitrile Butadiene Styrene (ABS)	N/A	N/A	None	None	Solid

Section 4 – Flammability Data

Components	Flashpoint	Explosive Limit	Comments
Lead and Sulfuric Acid	None	None	None
Hydrogen		LEL = 4.1%	Sealed batteries can emit hydrogen if overcharged (float voltage > 2.40 VPC)
Fiberglass Separator	N/A	N/A	Toxic vapors may be released. In case of fire, wear self contained breathing apparatus
Acrylonitrile Butadiene Styrene (ABS)	None	N/A	Temp over 527°F (300°C) may release combustible gases. In case of fire, wear self contained breathing apparatus

Section 5 - Reactivity Data

Stability	Unstable		Conditions to Avoid
	Stable	X	Prolonged overcharge on high current, ignition sources. Sulfuric acid remains stable at all temperatures
Incompatibility (Materials to Avoid)			
Sulfuric acid: Contact with combustibles and organic materials may cause fire and explosion. Also reacts violently with strong reducing agents, metals, sulfur trioxide gas, strong oxidizers, and water. Contact with metals may product toxic sulfur dioxide fumes and may release flammable hydrogen gas. Lead Compounds: Avoid contact with strong acids, bases, halides, halogenates, potassium nitrate, permanganate, peroxides, nascent hydrogen, and reducing agents.			
Hazardous Decomposition or Byproducts			
Sulfuric acid: Sulfur trioxide, carbon monoxide, sulfuric acid mist, sulfur dioxide, and hydrogen sulfide. Lead Compounds: High temperatures above the melting point are likely to produce toxic metal fume, vapor, or dust; contact with strong acid or base or presence of nascent hydrogen may generate highly toxic arsine gas. Hazardous Polymerization.			
Polymerization: Sulfuric acid will not polymerize			
Decomposition Products: Sulfuric Dioxide, Trioxide, Hydrogen Sulfide, Hydrogen.			
Conditions to Avoid: Prohibit smoking, sparks, etc. from battery charging area. Avoid mixing acid with other chemicals.			

Section 6 - Health Hazard Data

Routes of Entry
Sulfuric acid: Harmful by all routes of entry Lead compounds: Hazardous Exposure can occur only when product is heated, oxidized, or otherwise processed or damaged to create dust, vapor or fume.
Inhalation
Sulfuric Acid: Breathing sulfuric acid vapors and mists may cause severe respiratory problems. Lead Compounds: Dust or fumes may cause irritation of upper respiratory tract or lungs. Fiberglass Separator: Fiberglass is an irritant to the upper respiratory tract, skin and eyes. For exposure up to 10F°/ use MSA Comfoll with type H filter. Above 10F use Ultra Twin with type H filter. This product is not considered carcinogenic by NTP or OSHA.
Skin Contact
Sulfuric acid: Severe irritation, burns, cornea damage, and possible blindness. Lead Compounds: May cause eye irritation

Continued on next page

Power-Sonic MSDS Page 3 of 6

Ingestion Sulfuric acid: May cause severe irritation of the mouth, throat, esophagus, and stomach. Lead Compounds: May cause abdominal pain, nausea, vomiting, diarrhea, and severe cramping. Acute ingestion should be treated by a physician.
Eye Contact Sulfuric acid: Severe irritation, burns, cornea damage and possible blindness. Lead Compounds: May cause eye irritation.
Acute Health Hazards Sulfuric acid: Severe skin irritation, burns, damage to cornea may cause blindness, upper respiratory irritation. Lead Compounds: May cause abdominal pain, nausea, headaches, vomiting, loss of appetite, severe cramping, muscular aches and weakness, and difficulty sleeping. The toxic effects of lead are cumulative and slow to appear. It affects the kidneys, reproductive and central nervous systems. The symptoms of lead overexposure are listed above. Exposure to lead from a battery most often occurs during lead reclamation operations through the breathing or ingestion of lead dust or fumes.
Chronic Health Hazards Sulfuric acid: Possible scarring of the cornea, inflammation of the nose, throat and bronchial tubes, possible erosion of tooth enamel. Lead Compounds: May cause anemia, damage to kidneys and nervous system, and damage to reproductive system in both males and females.
Carcinogenicity Sulfuric acid: The National Toxicological Program (NTP) and The International Agency for Research on Cancer (IARC) have classified strong inorganic acid mist containing sulfuric acid as a Category 1 carcinogen, a substance that is carcinogenic to humans. The ACGIH has classified strong inorganic acid mist containing sulfuric acid as an A2 carcinogen (suspected human carcinogen). These classifications do not apply to liquid forms of sulfuric acid or sulfuric acid solutions contained within a battery. Inorganic acid mist (sulfuric acid mist) is not generated under normal use of this product. Misuse of the product, such as overcharging, may result in the generation of sulfuric acid mist. Lead Compounds: Human studies are inconclusive regarding lead exposure and an increased cancer risk. The EPA and the International Agency for Research on Cancer (IARC) have categorized lead and inorganic lead compounds as a B2 classification (probable/possible human carcinogen) based on sufficient animal evidence and inadequate human evidence.
Medical Conditions Generally Aggravated by Exposure Inorganic lead and its compounds can aggravate chronic forms of kidney, liver, and neurological diseases. Contact of battery electrolyte (acid) with the skin may aggravate skin diseases such as eczema and contact dermatitis. Overexposure to sulfuric acid mist may cause lung damage and aggravate pulmonary conditions.
Emergency and First Aid Procedures <u>Inhalation</u> Sulfuric acid: Remove to fresh air immediately. If breathing is difficult, give oxygen Lead Compounds: Remove from exposure, gargle, wash nose and lips, consult physician <u>Ingestion</u> Sulfuric acid: Do not induce vomiting, consult a physician immediately. Lead Compounds: Consult a physician immediately <u>Eyes</u> Sulfuric acid: Flush immediately with water for 15 minutes, consult a physician. Lead Compounds: Flush immediately with water for 15 minutes, consult a physician <u>Skin</u> Sulfuric acid: Flush with large amounts of water for at least 15 minutes, remove any contaminated clothing. If irritation develops seek medical attention. Lead Compounds: Wash with soap and water.

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Section 7 - Precautions for Safe Handling and Use

Steps to be Taken in Case Material is Released or Spilled There is no release of material unless the case is damaged or battery is misused/overcharged. If release occurs stop flow of material, contain/absorb all spills with dry sand, earth, or vermiculite. Do not use combustible materials. Neutralize spilled material with soda ash, sodium bicarbonate, lime, etc. Wear acid-resistant clothing, boots, gloves, and face shield. Dispose of as hazardous waste. Do not discharge acid to sewer
Waste Disposal Method Spent Batteries - send to secondary lead smelter for recycling. Follow applicable federal, state and local regulations. Neutralize as in preceding step. Collect neutralized material in sealed container and handle as hazardous waste as applicable. A copy of this MSDS must be supplied to any scrap dealer or secondary lead smelter with the battery.
Precautions to be Taken in Handling and Storing Store batteries in a cool, dry, well ventilated area that are separated from incompatible materials and any activities which may generate flames, sparks, or heat. Keep all metallic articles that could contact the negative and positive terminals on a battery and create a short circuit condition.
Electrical Safety Due to the battery's low internal resistance and high power density, high levels of short circuit current can be developed across the battery terminals. Do not rest tools or cables on the battery. Use insulated tools only. Follow all installation instructions and diagrams when installing or maintaining battery systems.
Fiberglass Separator Fiberglass is an irritant to the upper respiratory tract, skin and eyes. For exposure up to 10F% use MSA Comfoll with type H filter. Above 10F use Ultra Twin with type H filter. This product is not considered carcinogenic by NTP or OSHA.

Section 8 - Control Measures

Respiratory Protection None required under normal conditions. If battery is overcharged and concentrations of sulfuric acid are known to exceed PEL use NIOSH or MSH approved respiratory protection.	
Engineering Controls Store and handle batteries in a well ventilated area. If mechanical ventilation is used, components must be acid resistant	
Protective Gloves None needed under normal conditions. If battery case is damaged use rubber or plastic elbow length gauntlets	Eye Protection None needed under normal conditions. If handling damaged or broken batteries use chemical splash goggles or face shield
Other Protective Clothing or Equipment None needed under normal conditions. In case of damaged or broken battery use an acid resistant apron. Under severe exposure or emergency conditions wear acid resistant clothing.	
Work Hygienic Practices Handle batteries carefully to avoid damaging the case. Do not allow metallic articles to contact the battery terminals during handling. Avoid contact with the internal components of the battery.	

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4-4 General Battery and Charger Specifications

Refer to Tables 4-1 and 4-2 for manufacturer's general specifications for the POWERSONIC PS1270 battery and the EDS 11027 charger.

TABLE 4-1
GENERAL BATTERY SPECIFICATIONS

Power Sonic Model #	Nominal Voltage (V)	Nominal Capacity 20 hr rate (mA)	Discharge Current 20 hr rate (mA)	Dimensions								Approx. Weight		Standard Terminal Type
				Length		Width		Height		H.O.T.				
				in.	mm.	in.	mm.	in.	mm.	in.	mm.	lbs.	kg.	
PS1270	12	7.0	350	5.95	151	2.56	65	3.7	94	3.86	98	5.7	2.6	T2

GENERAL CHARGER SPECIFICATIONS

TABLE 4-2

EDS Model#	Nom. Voltage (V)	Voltage Range (V)	Nominal Current (A)	Type of Charger
11027	12	13.6- 14.70	0.70	Dual Voltage Auto

Charger Dimensions							
Length		Width		Height		Approx. Weight	
Inches	mm	inches	mm	Inches	mm	Lbs	Kg
3.4	86.4	2.8	71.1	2.3	58.42	1.3	0.59

Battery Recycling

Below is the web address for Environmental Safety and Health on-line. Enter this web address. Scroll down to "Where to Recycle Your Batteries". Here you will find two links to locate local recycling centers in California or all other states and Provinces in the US and Canada.

<http://www.ehso.com/ehshome/batteries.php>

5 PARTS LIST

5-1 Renewable Parts

See Table 5-1 for renewable parts for the DISCOVERY MR750 Main Disconnect Panel.

Table 5-1
LIST FOR RENEWABLE PARTS

Part	Volts	Amps	Buss	Gould-Shawmut	GEMS
Fuse FU1	380,480	1.5	FNQ-R-1-1/2	ATQR1-1/2	5161287-2
Fuse FU2	480	200	LPJ200SP	AJT-200	2375862-2
Fuse FU3	480	60	LPJ60SP	AJT-60	5161287-3
Red PB Operator*			CR104PBG01R1		
Contact Block*			P9B01VN		
Name Plate*			CR104PXN1RP088 "Emergency Off"		
SS Wall Plate*			97364		
Green PB Operator			P9SPNVG		5161287-4
Pilot Lamp (Green)			P9PDNVJLV		
Battery	12 DC	7.0Ah	PS1270		2276547-14
Battery Charger	12 DC	700ma	EDS 11027		2276547-16
* = Component parts of additional remote emergency OFF pushbutton					

All of the above parts are available from GEXPRO (Formally GE Supply) Milwaukee (414)-527-6600 or 1-800-200-9760.

5-2 Replacement Parts

See Table 5-2 for replacement parts for the DISCOVERY MR750 System Main Disconnect Panel.

Table 5-2
LIST FOR REPLACEMENT PARTS

Part	Manufacturer Part Number		
Power Distribution Lug Kit - CB1	TCAL28PD3		
Neutral Block	PDB371-1	CPDB-1	
CB1 – 200A, Lugs	SFHA36AT0250	(3) TCAL29	
200 AMP Plug – CB1, Line Lugs	SRPF250A200	TCAL28PD3	
CB2 – 150A, Lugs	SEHA36AT0150	(3) TCAL18	
150 AMP Plug – CB2, Padlock provision	SRPE150A150	SEPLD	
120V under voltage trip – CB2	SAUV1		
CB3 – 50A, Lugs	SEHA36AT0060	(6) TCAL18	
50 AMP Plug – CB3, padlock provision	SRPE60A50	SEPLD	
FU1 Fuse Holder, Fuse	CHCC2D	FNQR1.5	
FU2 Fuse Holder, Fuse	J60200-3CR	LPJ200SP	
FU2 Front Cover - Lexan	SD-451		
FU3 Fuse Holder, Fuse	JT60060	LPJ60SP	
CB4 – 3A	EP101ULC03		
CB5 – 1A	EP101ULC01		
Ground Lug & Ground Bus	(2) ADR25-21	TGL2	
C1 Contactor	CL06A311MJ		
T1 Transformer	9T58E0506		
CR1, Relay & socket 12VDC	IDEC RY4S-UDC12V	SOCKET SY4S-05, CLIP SY4S-51F1	
CR2 Relay 12VDC	451WFPR1D01		
Charger, 700mA	EDS 11027		
Battery, 7.2Ah, 12VDC	PS1270		
System Power “ON” PB’s (PB1)	P9SPNVG	P9B10VN, P9ASBGV028	
Red cover mounted Emergency OFF (PB2)	P9SPNVG	P9B01VN, P9ASBR029	
LT1 green Power On light and Power Supply	P9PDNVJLV	P9SLVD	FRU 5161287 -5
Remote Emergency Off pushbuttons	(2) CR104PBG01R1	(2) CR104PXN1RP088 , 97364	
CB1 Operating mechanism	SCH1	SCOM1EF	SC4L

Revision history

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
1	9/15/2008	L Hernandez	Editorial revisions, added FRU numbers